

MySQL

Exercise 1

1. Write a program that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, $\text{Perimeter} = 2*(L+W)$ and $\text{Area} = L*W$.)

Output-

```
mysql> create table expl(  
-> length float(7,2),  
-> width float(7,2)  
-> );  
Query OK, 0 rows affected, 2 warnings (0.23 sec)
```

```
mysql> delimiter //  
mysql> create procedure expone()  
-> Begin  
-> declare l float(7,2) default 12;  
-> declare w float(7,2) default 15;  
-> declare area float(7,2);  
-> declare perimeter float(7,2);  
->  
-> set perimeter = 2*(l+w);  
-> set area = l*w;  
->  
-> insert into expl values(area,perimeter);  
-> end //  
Query OK, 0 rows affected, 4 warnings (0.19 sec)
```

```
mysql> delimiter ;  
mysql> call expone();  
Query OK, 1 row affected (0.18 sec)
```

```
mysql> select*from expl;  
+-----+-----+  
| length | width |  
+-----+-----+  
| 180.00 | 54.00 |  
+-----+-----+  
1 row in set (0.01 sec)
```

2. Write a program that declares an integer variable called *num*, assigns a value to it, and computes and inserts into the *tempp* table the value of the variable itself, its square, and its cube.

Output-

```
mysql> delimiter //
mysql>
mysql> create procedure exptwop()
-> begin
->     declare x int;
->     declare square int(4);
->     declare cube_val int(4);
->
->     set x = 5;
->     set square = x * x;
->     set cube_val = x * x * x;
->
->     insert into tempp values(square, 'square');
->     insert into tempp values(cube_val, 'cube');
-> end //
Query OK, 0 rows affected, 2 warnings (0.22 sec)

mysql>
mysql> delimiter ;
mysql> call exptwop;
```

// 884 characters Plain text			
	25	square	
	125	cube	

3. Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are:- $C = (F - 32) * 5/9$
 $F = 9/5 * C + 32$

```

mysql> delimiter //
mysql> create procedure expthree()
    -> begin
    -> declare f float(7,2) default 12;
    -> declare c float(7,2) default 29;
    -> declare Fahrenheit float(7,2);
    -> declare Celsius float(7,2);
    ->
    -> set Fahrenheit = 9/5*C + 32 ;
    -> set Celsius = (F-32)*5/9 ;
    ->
    -> insert into expl values(Fahrenheit,Celsius);
    -> end //
Query OK, 0 rows affected, 4 warnings (0.04 sec)

mysql> delimiter ;
mysql> call expthree();
Query OK, 1 row affected (0.02 sec)

mysql> select*from expl;
+-----+-----+
| length | width  |
+-----+-----+
| 180.00 | 54.00  |
| 84.20  | -11.11 |
+-----+-----+
2 rows in set (0.00 sec)

mysql> |

```

3. Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

Output-

```
mysql> delimiter //
mysql> create procedure expfour()
  -> begin
  -> declare total_inches int default 124;
  -> declare yards int;
  -> declare feet int;
  -> declare inches int;
  -> declare rem int;
  ->
  -> set yards = total_inches/36;
  -> set rem = total_inches%36;
  ->
  -> set feet = rem/12;
  -> set inches = rem%12;
  ->
  -> insert into temp values(yards,'yards');
  -> insert into temp values(feet,'feet');
  -> insert into temp values(inches,'inches');
  -> end //
Query OK, 0 rows affected (0.06 sec)
```

3	yards
1	feet
4	inches

50 rows in set (0.00 sec)

- Write a program that enables a user to input an integer. The program should then state whether the integer is evenly divisible by 5.

Output-

```
mysql> delimiter //
mysql> create procedure expfiveone()
  -> begin
  -> declare x int(4);
  -> select sal into x from emp where ename ='KING';
  -> if (x%5=0) then
  -> insert into temp values(x,'integer is evenly divisible by 5');
  -> else
  -> insert into temp values(x,' not integer is evenly divisible by 5');
  ->
  -> end if ;
  -> end //
Query OK, 0 rows affected, 1 warning (0.11 sec)

mysql> delimiter ;
mysql>
mysql>
mysql> call expfiveone();
Query OK, 1 row affected (0.01 sec)
```

```
| 3000 | integer is evenly divisible by 5 |
+-----+-----+
51 rows in set (0.00 sec)
```

6. Your block should read in two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
mysql> delimiter //
mysql> create procedure expsix()
-> begin
-> declare x float default 25;
-> declare y float default 5;
-> declare product float;
->
-> set product = x * y;
->
-> if (product=100) then
-> insert into tempp values(x,'number is equal');
-> elseif (product>100) then
-> insert into tempp values(x,'number is greater');
-> else
-> insert into tempp values(x,'number is not equal and greater');
->
->
-> end if ;
-> end //
Query OK, 0 rows affected (0.58 sec)

mysql> delimiter ;
mysql> call expsix();
Query OK, 1 row affected (0.34 sec)
```

```
| 25 | number is greater |
+-----+-----+
1 row in set (0.00 sec)
```